

Assignment 3

Name: 项 枫 Student ID: 2022211570 Class Number: 2022211801

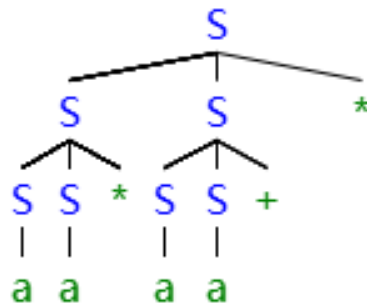
1 Required Exercises

Exercise 1

1、 No, it isn't.

$$2, \quad S \underset{lm}{\Rightarrow} SS \underset{lm}{\overset{*}{\Rightarrow}} SS * S \underset{lm}{\overset{*}{\Rightarrow}} aS * S \underset{lm}{\overset{*}{\Rightarrow}} aa * S \underset{lm}{\overset{*}{\Rightarrow}} aa * SS \underset{lm}{\overset{+}{\Rightarrow}} aa * aS \underset{lm}{\overset{+}{\Rightarrow}} aa * aa \underset{lm}{\overset{+}{\Rightarrow}}$$
$$3. \quad S \underset{rm}{\Rightarrow} SS \underset{rm}{\Rightarrow} SSS \underset{rm}{+*} SSa \underset{rm}{+*} Saa \underset{rm}{+*} SS \underset{rm}{+*} aa \underset{rm}{+*} Sa \underset{rm}{+*} aa \underset{rm}{+*} aa \underset{rm}{+*}$$

4、



5、 $S \rightarrow aS', S' \rightarrow S + S' | S * S' | \varepsilon.$

Exercise 2

1、

(a)

$$FIRST(S) = \{a\},$$

$$FIRST(B) = FIRST(S) \cup \{\varepsilon\} = \{a, \varepsilon\},$$

$$FOLLOW(S) = \{+, \$\},$$

$$FOLLOW(B) = FOLLOW(S) \cup \{\$ \} = \{+, \$\}.$$

(b)

NON-TERMINAL	INPUT SYMBOL		
	a	+	\$
S	$S \rightarrow aB$		
B	$B \rightarrow S + B$	$B \rightarrow \varepsilon$	$B \rightarrow \varepsilon$

2、 Yes, it is.

3、 Yes, it can.

MATCHED	STACK	INPUT	ACTION
	$S\$$	$aaaa+++\$$	
	$aB\$$	$aaaa+++\$$	$output\ S \rightarrow aB$
a	$B\$$	$aaa+++\$$	$match\ a$
a	$S+B\$$	$aaa+++\$$	$output\ B \rightarrow S+B$
a	$aB+B\$$	$aaa+++\$$	$output\ S \rightarrow aB$
aa	$B+B\$$	$aa+++\$$	$match\ a$
aa	$S+B+B\$$	$aa+++\$$	$output\ B \rightarrow S+B$
aa	$aB+B+B\$$	$aa+++\$$	$output\ S \rightarrow aB$
aaa	$B+B+B\$$	$a+++\$$	$match\ a$
aaa	$S+B+B+B\$$	$a+++\$$	$output\ B \rightarrow S+B$
aaa	$aB+B+B+B\$$	$a+++\$$	$output\ S \rightarrow aB$
$aaaa$	$B+B+B+B\$$	$+++\$$	$match\ a$
$aaaa$	$+B+B+B\$$	$+++\$$	$output\ B \rightarrow \varepsilon$
$aaaa+$	$B+B+B\$$	$++\$$	$match\ +$
$aaaa+$	$+B+B\$$	$++\$$	$output\ B \rightarrow \varepsilon$
$aaaa++$	$B+B\$$	$+\$$	$match\ +$
$aaaa++$	$+B\$$	$+\$$	$output\ B \rightarrow \varepsilon$
$aaaa+++$	$B\$$	$\$$	$match\ +$
$aaaa+++$	$\$$	$\$$	$output\ B \rightarrow \varepsilon$

2 Optional Exercises

Exercise 1

Yes, it is ambiguous.

e.g. 我喜欢你讨厌他

